# CS583 – Research Project Report
## SENTIMENT ANALYSIS

### 1. Introduction

**Project Overview:** The objective of this research project is to classify the sentiment or opinion expressed in tweets related to the presidential candidates Barack Obama and Mitt Romney into one of three classes: positive (1), negative (-1), or neutral (0). The training data consists of tweets about Obama and Romney, with class labels indicating positive, negative, neutral, or mixed sentiments. The test data, without class labels, will be used to evaluate the models.

### 2. Techniques Used

**Data Preprocessing:** The training data was loaded from an Excel file containing two sheets, one for Obama tweets and another for Romney tweets. The data underwent preprocessing to handle case normalization, punctuation, URL removal, and lemmatization. Additionally, tweets labeled as mixed (class 2) were excluded from training.

**Explanation of Data Preprocessing Code Logic:**

**Loading Data:** The **Load_Data** function loads data from an Excel file with specified sheet names, columns for tweets and labels, and additional parameters. Tweets with labels not in [-1, 0, 1] are filtered out, ensuring only relevant sentiment classes are considered. Missing tweet values are filled with "UNKNOWN." A new column **'Screen_Name'** is added to identify the source (Obama or Romney).

**Data Preprocessing Results:**

```
Rows with Screen_Name 'Obama':
  Screen_Name                                          Tweet  Label
0       Obama  Kirkpatrick, who wore a baseball cap embroider...      0
1       Obama  #<e>obama</e> debates that Cracker Ass Cracker...      1
2       Obama  @Hollivan @hereistheanswer  Youre missing the ...      0
3       Obama  I was raised as a Democrat  left the party yea...     -1
4       Obama  The <e>Obama camp</e> can't afford to lower ex...      0

Rows with Screen_Name 'Romney':
     Screen_Name                                            Tweet  Label
5471      Romney  Insidious!<e>Mitt Romney</e>'s Bain Helped Phi...     -1
5472      Romney  .@WardBrenda @shortwave8669 @allanbourdius you...     -1
5473      Romney  <e>Mitt Romney</e> still doesn't <a>believe</a...     -1
5474      Romney  <e>Romney</e>'s <a>tax plan</a> deserves a 2nd...     -1
5475      Romney  Hope <e>Romney</e> debate prepped w/ the same ...      1
```

**Text Processing:** The process function normalizes case, handles punctuation, and removes URLs from the raw text. The **process_all** function applies the process function to the **'Tweet'** column of the DataFrame, lemmatizing and tokenizing each tweet.

**Text Preprocessing Results:**

```
Rows with Screen_Name 'Obama':
  Screen_Name                                          Tweet  Label
0       Obama  [kirkpatrick, who, wear, a, baseball, cap, emb...      0
1       Obama  [e, obama, e, debate, that, cracker, as, crack...      1
2       Obama  [hollivan, hereistheanswer, youre, miss, the, ...      0
3       Obama  [i, be, raise, a, a, democrat, leave, the, par...     -1
4       Obama  [the, e, obama, camp, e, cant, afford, to, low...      0

Rows with Screen_Name 'Romney':
     Screen_Name                                            Tweet  Label
5471      Romney  [insidious, e, mitt, romney, e, bain, help, ph...     -1
5472      Romney  [wardbrenda, shortwave8669, allanbourdius, you...     -1
5473      Romney  [e, mitt, romney, e, still, doesnt, a, believe...     -1
5474      Romney  [e, romney, e, a, tax, plan, a, deserve, a, 2n...     -1
5475      Romney  [hope, e, romney, e, debate, prepped, w, the, ...      1
```

## 3. Model Evaluation:

Explored different techniques for sentiment analysis on political tweets using TF-IDF and Word2Vec representations. SVM emerged as top-performing model with accuracies of **59.62**% in second run. Certainly! Based on the result set, here's a description of the techniques tried, their results, and the lessons learned.

**Models Explored:**

1. **Naive Bayes:** Achieved an accuracy of **54.22%**.

```
Naive Bayes Model Results:
Testing Accuracy: 0.5422661870503597
Classification Report:
                 precision    recall  f1-score   support

           -1       0.53      0.85      0.65       927
            0       0.55      0.30      0.39       733
            1       0.59      0.35      0.44       564

     accuracy                           0.54      2224
    macro avg       0.56      0.50      0.49      2224
weighted avg        0.55      0.54      0.51      2224
```

2. **Logistic Regression:** Outperformed Naive Bayes with an accuracy of **56.8%**. Balanced precision and recall for all classes, resulting in a competitive performance.

```
Logistic Regression Model Results:
Testing Accuracy: 0.5683453237410072
Classification Report:
                 precision    recall  f1-score   support

           -1       0.59      0.73      0.65       927
            0       0.53      0.45      0.49       733
            1       0.56      0.46      0.51       564

     accuracy                           0.57      2224
    macro avg       0.56      0.55      0.55      2224
weighted avg        0.56      0.57      0.56      2224
```

3. **Support Vector Machines (SVM):** Achieved an accuracy of **58.22%**. Provided balanced precision and recall, indicating a robust model.

```
SVM Model Results:
Testing Accuracy: 0.5822841726618705
Classification Report:
                precision    recall  f1-score   support

          -1       0.57      0.79      0.66       927
           0       0.57      0.42      0.49       733
           1       0.63      0.45      0.52       564

    accuracy                           0.58      2224
   macro avg       0.59      0.55      0.56      2224
weighted avg       0.59      0.58      0.57      2224
```

4. **Multilayer Perceptron (MLP):** Achieved an accuracy of **53.8%**.

```
MLP Model Results:
Testing Accuracy: 0.5382194244604317
Classification Report:
                precision    recall  f1-score   support

          -1       0.58      0.62      0.60       927
           0       0.48      0.44      0.46       733
           1       0.52      0.53      0.53       564

    accuracy                           0.54      2224
   macro avg       0.53      0.53      0.53      2224
weighted avg       0.54      0.54      0.54      2224
```

5. **Random Forest:** Performed well with an accuracy of **57.4**%. Strong recall for class -1 but slightly lower accuracy compared to Logistic Regression and SVM.

```
Random Forest Model Results:
Testing Accuracy: 0.5741906474820144
Classification Report:
                precision    recall  f1-score   support

          -1       0.56      0.80      0.66       927
           0       0.55      0.42      0.48       733
           1       0.64      0.40      0.49       564

    accuracy                           0.57      2224
   macro avg       0.59      0.54      0.54      2224
weighted avg       0.58      0.57      0.56      2224
```

**Reasons for choosing SVM and TF-IDF:**

1. **Support Vector Machines (SVM):** SVM showed competitive accuracy (around 59-60%) compared to other models like Logistic Regression and Naive Bayes. SVM is less prone to overfitting, making it robust in scenarios where data size is limited. **RandomizedSearchCV** was employed to explore optimal hyperparameters for SVM, ensuring a thorough search across potential configurations.
2. **TF-IDF (Term Frequency-Inverse Document Frequency):** TF-IDF is effective in transforming text data into numerical vectors, capturing the importance of terms in documents. This prevents overfitting and enhances model

generalization. Unlike Word2Vec, which produces dense vector embeddings, TF-IDF vectors are sparse and retain the interpretability of individual terms.

**SVM Model Run 2:** SVM emerged as top-performing model with accuracies of **59.62**% in second run.

```
Best Parameters for SVC (RandomizedSearchCV - Run 2): {'classifier__kernel': 'rbf', 'classifier__C': 1}

SVC Model Results (RandomizedSearchCV - Run 2):
Accuracy: 0.5962230215827338

Classification Report:
              precision    recall  f1-score   support

          -1       0.59      0.83      0.69       927
           0       0.59      0.40      0.48       745
           1       0.63      0.48      0.54       552

    accuracy                           0.60      2224
   macro avg       0.60      0.57      0.57      2224
weighted avg       0.60      0.60      0.58      2224
```

**Lessons Learned:**

1. **Model Selection:** SVM emerged as top-performing models. Choice between models may depend on interpretability, training time, and scalability.
2. **Evaluation Metrics:** Balancing precision and recall are crucial for sentiment analysis, especially when dealing with imbalanced datasets.
3. **Feature Representation:** TF-IDF and Word2Vec complement each other; TF-IDF captures document-level information, while Word2Vec captures word semantics.
4. **Hyperparameter Tuning:** Further exploration of hyperparameter tuning could enhance model performance.
5. **Data Quality:** The performance of models is highly dependent on the quality and representativeness of the training data.
6. **Scalability:** Consideration of model scalability is essential, especially when dealing with large datasets.

**In conclusion**, the experimentation process involved a systematic exploration of various models and representations. The lessons learned emphasize the importance of model interpretability, balancing metrics, and continuous refinement through hyperparameter tuning and ensemble methods.

To further enhance accuracy and consider the benefits of deep learning models,

- **Deep Learning Architectures:** We need to consider models like **RNNs** or **BERT**, known for capturing complex patterns in sequential data, enhancing sentiment analysis by capturing nuanced expressions.
- **Transfer Learning:** We need to utilize pre-trained models like **GPT** or **BERT**, leveraging their language understanding for sentiment analysis. Fine-tuning on specific tasks transfers knowledge effectively, improving accuracy.